Clinical Study Summary Report (CSSR) Document Status: Draft | **Version:** 1.0 | **Date:** 03-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A Study to Evaluate the Safety, Tolerability, Pharmacokinetics, Pharmacodynamics, and Efficacy of NXT007 in Persons With Severe or Moderate Hemophilia A **Short Title/Acronym:** WP44714 / NXT007-01 **Protocol Number:** WP44714, 2023-503906-35-00 **NCT Number:** NCT05987449 **Sponsor Name:** Hoffmann-La Roche **Investigational Product:** NXT007 (next-generation bispecific antibody / FVIIIa-mimetic) **Phase of Development:** Phase I/II **Medical Monitor:** [Name and Contact Information]

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To investigate the safety and tolerability of NXT007, including the incidence, severity, and relatedness of adverse events (AEs) and serious adverse events (SAEs). **Secondary Objectives:** To evaluate efficacy (annualized bleeding rate [ABR]), pharmacokinetics (PK), pharmacodynamics (PD, including Thrombin Generation), and immunogenicity (incidence of anti-drug antibodies [ADA]).

2.2 Background & Rationale Hemophilia A is a genetic disease caused by a missing or defective blood clotting protein (Factor VIII), resulting in spontaneous or prolonged bleeding, joint damage, and life-threatening hemorrhages. NXT007 is a next-generation bispecific antibody, based on the framework of emicizumab, designed to mimic the cofactor function of activated factor VIII. In comparison to emicizumab, NXT007 has demonstrated greater potency, a higher maximum effect in thrombin generation, and a longer half-life, with the goal of achieving non-hemophiliac levels of coagulation potential utilizing a user-friendly monthly subcutaneous dosing regimen.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Multiple-ascending-dose [MAD]) **Blinding:** Open-label **Randomization:** Non-randomized

3.2 Planned Enrollment & Duration Target N: Approximately 60 participants **Number of Sites:** Global, multicenter (including sites in the US, Canada, UK, Spain, Poland, etc.) **Estimated Study Start:** September 2023 **Estimated Primary Completion:** June 2030 (estimated overall completion)

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male participants aged 2 to 59 years (Part 1: adults/adolescents 12-59 years; Part 2: pediatric 2-11 years). 2. Confirmed diagnosis of severe (FVIII coagulant activity <1 IU/dL) or moderate (FVIII coagulant activity 1-5 IU/dL) congenital Hemophilia A. 3. Participants with or without FVIII inhibitors. 4. Adequate hepatic (e.g., total bilirubin $\le 1.5\times$ ULN, AST/ALT $\le 3\times$ ULN) and renal function.

4.2 Exclusion Criteria 1. Known HIV infection with CD4 counts < 200 cells/µL. 2. History of severe allergic or anaphylactic reactions to monoclonal antibody therapies, chimeric/humanized antibodies, or Chinese hamster ovary cell products. 3. Clinically significant ECG abnormalities (e.g., QTcF >450 ms, left bundle branch block, structural heart disease). 4. Recent use of investigational drugs (within 30 days or 5 half-lives), prior gene therapy, or systemic immunomodulators (e.g., interferon or rituximab).

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Multiple-ascending-dose (MAD). The first 2 loading doses are given on Days 1 and 15 (e.g., 0.42 mg/kg, 1.05 mg/kg, or 1.62 mg/kg depending on cohort). Maintenance doses are administered every 4 weeks starting on Day 29 (e.g., 0.28 mg/kg, 0.70 mg/kg, or 1.08 mg/kg depending on cohort). **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** The primary treatment phase is 24 weeks. Participants may continue NXT007 maintenance dosing for up to 7 years in the extension phase.

5.2 Concomitant & Prohibited Therapy Permitted: Anti-retroviral therapy to treat HIV; bypass agents such as rFVIIa for the treatment of breakthrough bleeds, trauma, or procedures. **Prohibited:** Other investigational drugs, gene therapies, and systemic immunomodulators.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Labs, ECG, FVIII Inhibitor Testing
Baseline	Day 1	First Loading Dose, Vitals, Baseline PK/PD
Interim 1	Day 15 ($\pm$ 2)	Second Loading Dose, Safety Labs, Efficacy Assessment
Maintenance	Day 29 and Q4W	Maintenance Doses Every 4 Weeks, Bleed Rate Tracking, PK/PD, ADA Monitoring
End of Study	Up to 7 Years	Final Safety Review, Exit Interview, IP Accountability

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence, severity, and relatedness of Adverse Events (AEs) and Serious Adverse Events (SAEs) over the study duration. **Measurement Method:** Clinical assessment evaluated per standard toxicity scales (e.g., WHO Toxicity Scale).

7.2 Secondary & Exploratory Endpoints * Annualized Bleeding Rate (ABR) assessed using the ISTH-SSC 72-hour rule. * Pharmacokinetics (PK) plasma concentrations of NXT007. * Pharmacodynamics (PD), specifically target engagement parameters including activated FXI-triggered Thrombin Generation (TG). * Immunogenicity (incidence of anti-drug antibodies [ADA]).

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring during clinic visits; tracking of injection-site reactions, thromboembolic events, and thrombotic microangiopathies. **Serious Adverse Events (SAEs):** Standard expedited reporting timeline (typically 24 hours of site awareness). **Data Monitoring Committee (DMC):** An independent/internal Safety Review Committee monitors safety/PK data across cohorts before escalating doses to subsequent groups.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants who receive ≥ 1 dose of study drug). **Sample Size Justification:** Approximately 60 participants overall, appropriately sized to meet safety, tolerability, PK, and PD evaluation requirements for a Phase I/II multiple-ascending-dose trial without formal power calculations for efficacy superiority. **Handling of Missing Data:** A negative binomial regression model will be utilized to account for variable follow-up durations when calculating the Annualized Bleeding Rate (ABR).

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring to ensure protocol adherence, drug accountability, and patient safety. **Audit Trail:** Data entered into validated eCRFs with standardized data clarification form (DCF) processes.

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11. Study Identifiers & Governance

Primary ID: WP44714 **Registry Number:** NCT05987449 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** NXT007-01 / WP44714 Phase I/II Global MAD Study **Official Regulatory Title:** A Study to Evaluate the Safety, Tolerability, Pharmacokinetics, Pharmacodynamics, and Efficacy of NXT007 in Persons With Severe or Moderate Hemophilia A

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12. Clinical Framework (Hemophilia A Specific)

Primary Condition: Hemophilia A **Diagnosis Threshold:** Severe (FVIII <1 IU/dL) or Moderate (FVIII 1-5 IU/dL) **Medical Methodology:** Prophylactic Coagulation Therapy **Optimization Target:** Restoration of intrinsic tenase activity / non-hemophiliac levels of coagulation potential

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13. Study Procedures & Methodology

Management Pathway: Scheduled subcutaneous prophylactic antibody administration **Education Protocol (HCP):** Dose escalation safety thresholds, bypassing agent guidelines for breakthrough bleeds **Education Protocol (Patient):** Subcutaneous injection site management, bleed diary logging **Quality Audit Mechanism:** Safety Review Committee oversight between cohort dose escalations

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14. Observation & Follow-Up Schedule

Total Duration: Up to 7 Years (Extension Phase) **Onsite Visit Cadence:** Days 1, 15, and 29, then Every 4 Weeks (Q4W) **Remote Monitoring Frequency:** Intermittent remote check-ins between Q4W maintenance visits **Checklist Review Interval:** Every 4 Weeks (Q4W) with clinical site visits

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	**[DD-MMM-YYYY]**
Clinical Operations Lead	**[Name]**	___	***[DD-MMM-YYYY]***
Lead Statistician	***[Name]***	_____	[DD-MMM-YYYY]